# Week 1 Internship Report

**Organization:** Cyberster
**Intern Role:** Red team member (Information Gathering Phase)
**Intern:** Ayesha Memon

## 1. Introduction

During the first week of my cybersecurity internship, I worked on the **Information Gathering (Reconnaissance) phase** of penetration testing. This phase focuses on collecting publicly available information about a target without directly interacting with or attacking its infrastructure.

The tasks involved **Passive OSINT, Subdomain Enumeration, Technology Profiling, GitHub Dorking, and Subdomain Validation** using multiple industry tools.

# Task 1 – Passive OSINT & Subdomain Enumeration

## Tools Used

### 1. Subfinder

Subfinder is a passive subdomain discovery tool that collects subdomains from **multiple OSINT sources such as DNS records, certificate logs, and search engines**.

### 2. Assetfinder

Assetfinder is another subdomain enumeration tool that gathers subdomains from **public internet sources and certificate transparency logs**.

### 3. Amass

Amass is a powerful reconnaissance tool used for **network mapping and subdomain discovery** using OSINT.

## Combining the Results

All discovered subdomains from the three tools were combined into a single file.

Example process:

```
cat subfinder.txt assetfinder.txt amass.txt > hacksubhack.txt
```

Total discovered subdomains:
**81**

```
Session   Actions   Edit   View   Help
┌──(kali㉿kali)-[~]
└─$ cat subhack.txt
v3dev.hackthissite.org
0-v3stage-cdn.hackthissite.org
1-v3dev-cdn.hackthissite.org
forums.hackthissite.org
h5ai.hackthissite.org
lille.irc-v6.hackthissite.org
3-v3dev-cdn.hackthissite.org
api.hackthissite.org
www.irc.hackthissite.org
hp.hackthissite.org
mirror.hackthissite.org
status.hackthissite.org
v3stage-cdn.hackthissite.org
htsv4.hackthissite.org
vm-099.outbound.firewall.hackthissite.org
v3stage.hackthissite.org
irc-wolf.hackthissite.org
status-new.hackthissite.org
0-v3dev-cdn.hackthissite.org
4-v3dev-cdn.hackthissite.org
email.hackthissite.org
irc-www.hackthissite.org
oi.hackthissite.org
ns2.hackthissite.org
irc-lille.hackthissite.org
git.hackthissite.org
lille.irc.hackthissite.org
2-v3dev-cdn.hackthissite.org
ctf.hackthissite.org
qdb.hackthissite.org
vm-150.outbound.firewall.hackthissite.org
irc-v6.hackthissite.org
```

```
ns1.hackthissite.org
ns2.hackthissite.org
numex/
numex/numex.dat
pi.hackthissite.org
psrp.hackthissite.org
qdb.hackthissite.org
shadow.hackthissite.org
staff.hackthissite.org
stats.hackthissite.org
status.hackthissite.org
status-new.hackthissite.org
transliteration/
transliteration/transliteration.dat
v3dev-cdn.hackthissite.org
v3dev.hackthissite.org
v3stage-cdn.hackthissite.org
v3stage.hackthissite.org
vm-005.outbound.firewall.hackthissite.org
vm-050.outbound.firewall.hackthissite.org
vm-099.outbound.firewall.hackthissite.org
vm-150.outbound.firewall.hackthissite.org
vm-200.outbound.firewall.hackthissite.org
wolf.irc.hackthissite.org
wolf.irc-v6.hackthissite.org
www.hackthissite.org
www.irc.hackthissite.org

┌──(kali㉿kali)-[~]
└─$ wc -l final_subdomain.txt
81 final_subdomain.txt
```

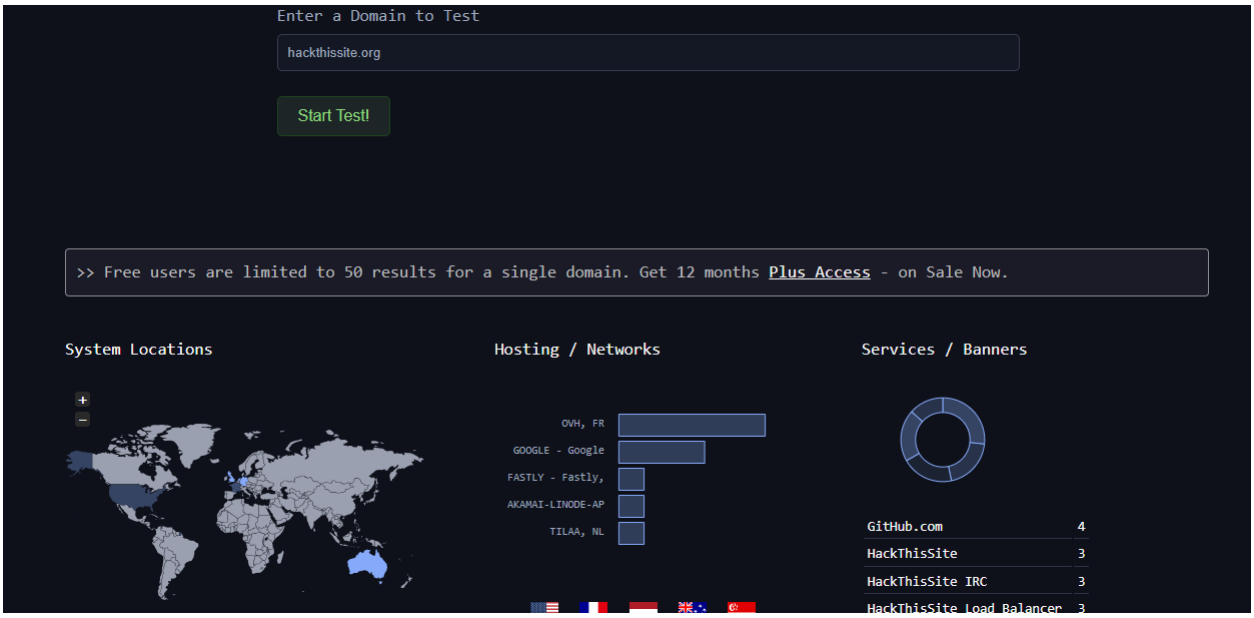
## Additional OSINT Platforms

### DNSDumpster

DNSDumpster is a web-based tool used to analyze **DNS records and infrastructure of a domain**.

Information discovered:

- DNS records
- Host information
- Subdomains




| Name | Count |
|---|---|
| GitHub.com | 4 |
| HackThisSite | 3 |
| HackThisSite IRC | 3 |
| HackThisSite Load Bala | 3 |
| nginx/1.18.0 (Ubuntu) | 2 |

| Top 5 ASN | | | |
|---|---|---|---|
| ASN | ASN Name | Network Range | Count |
| 16276 | OVH, FR | 137.74.0.0/16 | 12 |
| 15169 | GOOGLE - Google LLC | 172.253.116.0/24 | 7 |
| 54113 | FASTLY - Fastly, Inc. | 185.199.111.0/24 | 2 |
| 63949 | AKAMAI-LINODE-AP Akamai Connected Cloud, SG | 172.239.32.0/19 | 2 |
| 196752 | TILAA, NL | 185.24.220.0/22 | 2 |

| Top 5 Countries | |
|---|---|
| Name | Count |
| United States | 13 |
| France | 12 |
| Netherlands | 3 |
| Australia | 1 |
| Singapore | 1 |

## crt.sh

crt.sh is a **certificate transparency search engine** that allows users to find subdomains by analyzing SSL certificate logs.

crt.sh Identity Search Group by Issuer

| Criteria | Type: Identity Match: ILIKE Search: 'hackthissite.org' |
|---|---|

ificates

| crt.sh ID | Logged At | Not Before | Not After | Common Name | Matching Identities | Issuer Name |
|---|---|---|---|---|---|---|
| 24709094335 | 2026-03-01 | 2026-03-01 | 2026-05-30 | lille.irc.hackthissite.org | irc.hackthissite.org<br>irc-v6.hackthissite.org<br>lille.irc.hackthissite.org<br>lille.irc-v6.hackthissite.org | C=US, O=Let's Encrypt, CN=R12 |
| 24709093493 | 2026-03-01 | 2026-03-01 | 2026-05-30 | lille.irc.hackthissite.org | irc.hackthissite.org<br>irc-v6.hackthissite.org<br>lille.irc.hackthissite.org<br>lille.irc-v6.hackthissite.org | C=US, O=Let's Encrypt, CN=R12 |
| 24668715572 | 2026-02-27 | 2026-02-27 | 2026-05-28 | ctf.hackthissite.org | ctf.hackthissite.org | C=US, O=Let's Encrypt, CN=R13 |
| 24668693220 | 2026-02-27 | 2026-02-27 | 2026-05-28 | ctf.hackthissite.org | ctf.hackthissite.org | C=US, O=Let's Encrypt, CN=R13 |
| 24668110978 | 2026-02-27 | 2026-02-27 | 2026-05-28 | hackthissite.org | *.hackthissite.org<br>hackthissite.org | C=US, O=Let's Encrypt, CN=R13 |
| 24668103264 | 2026-02-27 | 2026-02-27 | 2026-05-28 | hackthissite.org | *.hackthissite.org<br>hackthissite.org | C=US, O=Let's Encrypt, CN=R13 |
| 24648823315 | 2026-02-26 | 2026-02-26 | 2026-05-27 | hackthissite.org | *.hackthissite.org<br>hackthissite.org | C=US, O=Let's Encrypt, CN=E7 |
| 24648822224 | 2026-02-26 | 2026-02-26 | 2026-05-27 | hackthissite.org | *.hackthissite.org<br>hackthissite.org | C=US, O=Let's Encrypt, CN=E7 |
| 24531059147 | 2026-02-20 | 2026-02-20 | 2026-05-21 | h5ai.hackthissite.org | h5ai.hackthissite.org | C=US, O=Let's Encrypt, CN=R13 |
| 24531035607 | 2026-02-20 | 2026-02-20 | 2026-05-21 | h5ai.hackthissite.org | h5ai.hackthissite.org | C=US, O=Let's Encrypt, CN=R13 |
| 24213665858 | 2026-02-06 | 2026-02-06 | 2026-05-07 | wolf.irc.hackthissite.org | irc.hackthissite.org<br>irc-v6.hackthissite.org<br>wolf.irc.hackthissite.org<br>wolf.irc-v6.hackthissite.org | C=US, O=Let's Encrypt, CN=R12 |

# Task 2 – Technology Profiling

## What is Technology Profiling?

Technology profiling identifies the **technologies, frameworks, and infrastructure used by a website**.

This helps penetration testers understand:

- Operating system
- Web server
- Programming languages
- JavaScript frameworks
- IP addresses

## tool Used: WhatWeb

WhatWeb is a reconnaissance tool that identifies **web technologies used by a website**.

## Information Discovered

The tool revealed several technologies including:

- Server IP address
- Operating System: Ubuntu
- Web technologies: HTML, JavaScript
- Server location: United States / Canada
- Additional infrastructure details

```
┌──(kali㉿kali)-[~]
└─$ cat whatweb_report.txt
http://4-v3dev-cdn.hackthissite.org/ [400 Bad Request] Content-Language[en], Country[UNITED STATES][US], Email[webmaster], HTTPServer[squid], IP[198.148.81.145], mayb
e Squid-Web-Proxy-Cache, Title[ERROR: The requested URL could not be retrieved], UncommonHeaders[x-squid-error], X-Cache[serv,serv:8800]
http://0-v3stage-cdn.hackthissite.org/ [400 Bad Request] Content-Language[en], Country[UNITED STATES][US], Email[webmaster], HTTPServer[squid], IP[198.148.81.146], ma
ybe Squid-Web-Proxy-Cache, Title[ERROR: The requested URL could not be retrieved], UncommonHeaders[x-squid-error], X-Cache[serv,serv:8800]
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http://0-v3dev-cdn.hackthissite.org/ [400 Bad Request] Content-Language[en], Country[UNITED STATES][US], Email[webmaster], HTTPServer[squid], IP[198.148.81.145], mayb
e Squid-Web-Proxy-Cache, Title[ERROR: The requested URL could not be retrieved], UncommonHeaders[x-squid-error], X-Cache[serv,serv:8800]
http://ctf.hackthissite.org [301 Moved Permanently] HTTPServer[GitHub.com], IP[185.199.111.153], RedirectLocation[https://ctf.hackthissite.org/], Title[301 Moved Perm
anently], UncommonHeaders[x-github-request-id,x-served-by,x-cache-hits,x-timer,x-fastly-request-id], Via-Proxy[1.1 varnish]
http://irc.hackthissite.org [301 Moved Permanently] Country[CANADA][CA], HTTPServer[Ubuntu Linux][nginx/1.18.0 (Ubuntu)], IP[137.74.187.150], RedirectLocation[https:/
/www.irc.hackthissite.org/], Strict-Transport-Security[max-age=63072000; includeSubDomains; preload], Title[301 Moved Permanently], UncommonHeaders[x-content-type-opt
ions], X-Frame-Options[DENY], nginx[1.18.0]
http://3-v3dev-cdn.hackthissite.org/ [400 Bad Request] Content-Language[en], Country[UNITED STATES][US], Email[webmaster], HTTPServer[squid], IP[198.148.81.145], mayb
e Squid-Web-Proxy-Cache, Title[ERROR: The requested URL could not be retrieved], UncommonHeaders[x-squid-error], X-Cache[serv,serv:8800]
http://4-v3stage-cdn.hackthissite.org/ [400 Bad Request] Content-Language[en], Country[UNITED STATES][US], Email[webmaster], HTTPServer[squid], IP[198.148.81.146], ma
ybe Squid-Web-Proxy-Cache, Title[ERROR: The requested URL could not be retrieved], UncommonHeaders[x-squid-error], X-Cache[serv,serv:8800]
http://h5ai.hackthissite.org [301 Moved Permanently] HTTPServer[GitHub.com], IP[185.199.110.153], RedirectLocation[https://h5ai.hackthissite.org/], Title[301 Moved Pe
rmanently], UncommonHeaders[x-github-request-id,x-served-by,x-cache-hits,x-timer,x-fastly-request-id], Via-Proxy[1.1 varnish]
http://irc-hub.hackthissite.org/ [400 Bad Request] Content-Language[en], Country[UNITED STATES][US], Email[webmaster], HTTPServer[squid], IP[198.148.81.169], maybe Sq
uid-Web-Proxy-Cache, Title[ERROR: The requested URL could not be retrieved], UncommonHeaders[x-squid-error], X-Cache[serv,serv:8800]
http://3-v3stage-cdn.hackthissite.org/ [400 Bad Request] Content-Language[en], Country[UNITED STATES][US], Email[webmaster], HTTPServer[squid], IP[198.148.81.146], ma
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http://1-v3stage-cdn.hackthissite.org/ [400 Bad Request] Content-Language[en], Country[UNITED STATES][US], Email[webmaster], HTTPServer[squid], IP[198.148.81.146], ma
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http://1-v3dev-cdn.hackthissite.org/ [400 Bad Request] Content-Language[en], Country[UNITED STATES][US], Email[webmaster], HTTPServer[squid], IP[198.148.81.145], mayb
e Squid-Web-Proxy-Cache, Title[ERROR: The requested URL could not be retrieved], UncommonHeaders[x-squid-error], X-Cache[serv,serv:8800]
http://2-v3stage-cdn.hackthissite.org/ [400 Bad Request] Content-Language[en], Country[UNITED STATES][US], Email[webmaster], HTTPServer[squid], IP[198.148.81.146], ma
ybe Squid-Web-Proxy-Cache, Title[ERROR: The requested URL could not be retrieved], UncommonHeaders[x-squid-error], X-Cache[serv,serv:8800]
http://hub.irc.hackthissite.org [400 Bad Request] Content-Language[en], Country[UNITED STATES][US], Email[webmaster], HTTPServer[squid], IP[198.148.81.169], maybe Squ
```

# Task 3 – GitHub Dorking

## What is GitHub Dorking?

GitHub Dorking is a technique used to search for **sensitive information accidentally exposed in public repositories**.

Developers sometimes upload configuration files containing:

- API keys
- Credentials
- Tokens
- Sensitive URLs

## Method Used

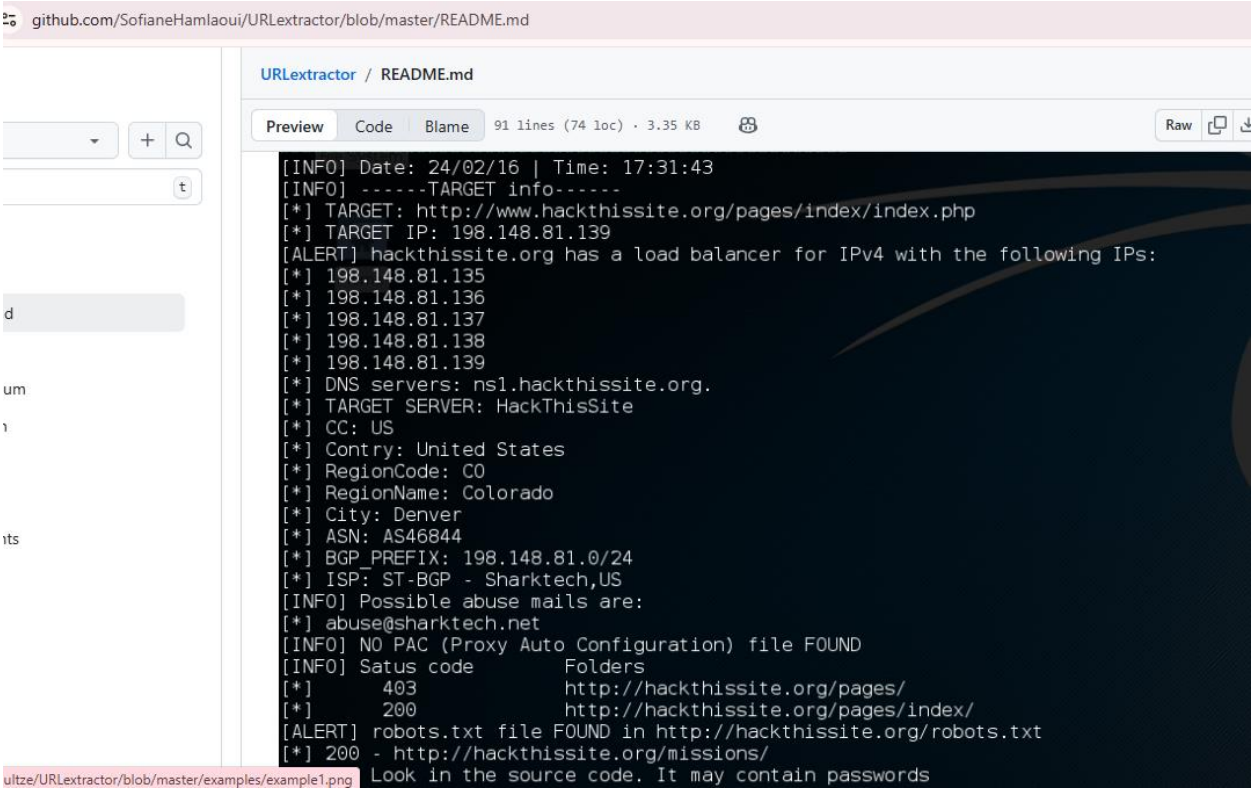
Using targeted search queries related to the domain, I performed manual GitHub searches to identify possible leaked information.

Example search patterns:

```
targetdomain.com password
targetdomain.com api_key
targetdomain.com secret
```

## Findings

During the investigation, some potential credentials and sensitive information were observed in public repositories.




# Task 4 – Alive Subdomain Detection

Not all discovered subdomains are active. Therefore, the next step was identifying which subdomains were **alive and responding to HTTP requests**.

# Tools Used

## HTTPProbe

HTTPProbe checks whether a domain or subdomain is responding to HTTP or HTTPS requests.

## HTTPX

HTTPX is a fast tool used for probing **HTTP services and identifying live hosts**.

# Final Results

Total subdomains discovered:
**81**

Alive subdomains:
**51**

All active subdomains were saved in a file:

`alive.txt`

```
Session   Actions   Edit   View   Help
http://lille.irc.hackthissite.org
http://hp.hackthissite.org
https://irc-lille.hackthissite.org
http://irc-lille.hackthissite.org
https://mirror.hackthissite.org
https://legal.hackthissite.org
http://mirror.hackthissite.org
http://irc-hub.hackthissite.org
http://legal.hackthissite.org
http://hub.irc.hackthissite.org
https://qdb.hackthissite.org
https://stats.hackthissite.org
https://status.hackthissite.org
http://qdb.hackthissite.org
http://stats.hackthissite.org
http://status.hackthissite.org
http://v3dev-cdn.hackthissite.org
http://v3dev.hackthissite.org
http://v3stage-cdn.hackthissite.org
http://v3stage.hackthissite.org
http://psrp.hackthissite.org
https://wolf.irc.hackthissite.org
http://wolf.irc.hackthissite.org
https://www.irc.hackthissite.org
http://www.irc.hackthissite.org
https://www.hackthissite.org
http://www.hackthissite.org

┌──(kali㉿kali)-[~]
└─$ wc -l alive.txt
51 alive.txt
```